July Assignment for ADP470S 2020.

Group Assignment (Max 5 in a group)

Weight 35%

Instructions:

1. Any two program per group. You are allowed to book any two. As discussed during class communicate in your WhatsApp group and class rep so not more than 4 groups can take the same combination.
2. Due date is (as per the discussion during class) midnight 26th July 2020.

Please choose the 2 out of the 6 problems given under the headline problems (page 2) what should you submit Via a PPT for each?

1. A title page (Mentioning Problem 1 / 2 / etc.) with specifying the Surname, Initial and Student Number of the participating members. (1)
2. Introduction to the problem (2)
3. A detailed algorithm on how you are approaching to solve the problem (30)
4. A small guideline on how a user should run and use your code. (2)
5. A small description of what you learnt from the assignment. (5)
6. A screenshot of the output while you were running the program clearly showing the input, output and time taken of the program. (20)
7. The source code in another word file so the examiner can copy paste and check the output as demanded in your PPT other than the time taken. The source code must come with comment lines and should adhere with the Algorithm stated in point 3. (15)

Total; 75 * 2 = 150

Please compress (.Zip / .Rar) all the submission and upload via the link in Black Board.

The lecturer reserves the right to call any group for a interview via BB Collaborate / Zoom / Teams for clarification regarding the submission. If found suspicious or found the same code in a book or internet the group will be marked zero or called for disciplinary via the department.

THE PROBLEMS:

1. Explain how to implement two stacks in one array called Arr of size n in such a way that neither stack overflows unless the total number of elements in both stacks together is n.

2. Show how to implement a queue using two stacks.

3. Show how to implement a stack using two queues.

4. Given two linked lists. Show how to find that whether the data in one is reverse that of data in another. No extra space should be used and traverse the linked lists only once.

5. Multiplication of two polynomials

   (e.g. $(5x^2+3x+2) * (7x + 1)$) using Linked list

6. To evaluate a postfix expression using a stack. Such as: 1 2 + 3 4 + *

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